Tasneea Zahra

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Education

Khulna University of Engineering & Technology (KUET) Jan 2020 – Sept 2025
BSc in Electrical and Electronic Engineering; Major: CSP

CGPA: 3.61/4.0

Relevant coursework: Digital Image Processing; Microprocessors; Numerical Methods and Statistics; Data Structures and Algorithms; Wireless Communication; Power Electronics and Industrial Drives; VLSI Design and Technology; Electrical Machines; Control Systems; Digital Electronics and Logic Design; Communication Engineering; Digital Signal Processing; Artificial Intelligence and Machine Learning; Pattern Recognition.

Thesis: Implementation and Performance Analysis of a Hybrid CNN–Transformer Model for Polyp Image Segmentation.

Research Focus

My research interests span reinforcement learning, algorithms, multimodal learning, and computer vision, with a focus on efficient LLM training. I am drawn to problems where better algorithms reduce the data and computation needed to train large models. I explore these ideas through policy optimization for language-model reasoning, vision-language models, and medical image analysis.

Publications & Preprints

[1] **Skip What You Can Predict: Predictive Repositioning for Policy Optimization for Efficient LLM Training**
I. N. Swapnil, A. Saha, **T. Zahra**, T. A. Khan, M. A. Haque, & S.-N. Lim

RLVR for language-model reasoning is costly because every same-batch optimizer step needs a full update. PrePO extrapolates from two observed steps to a farther point on the trajectory and corrects it, matching baseline performance with fewer optimization steps and less wall-clock time; the gains carry over to supervised fine-tuning.

Submitted to ICLR 2027 · Preprint · arXiv:2605.06755

[2] **Trans-SPAnet: A Unified Transformer–CNN Architecture with Enhanced Boundary Optimization for Polyp Segmentation**

A hybrid CNN–Transformer for polyp segmentation in colonoscopy images. Spatial-context preservation blocks retain fine detail, a Transformer decoder models long-range context, and boundary-aware optimization sharpens polyp contours, improving on CNN-only baselines.

IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON 2025)

Research Experience

University of Central Florida Nov 2025 – Present
Research Collaborator Orlando, FL (Remote)

- Collaborating with **Prof. Ser-Nam Lim** on efficient policy optimization for large language model reasoning.
- Co-author of **PrePO**: predictive repositioning approximates repeated same-batch optimizer updates at a fixed active cost, cutting explicit optimization steps and wall-clock time at matched performance in RL with verifiable rewards.
- Contributing to the empirical validation of the method on **language-model reasoning**, where the gains also carry over to **supervised fine-tuning**, and to the preparation of the paper for ICLR 2027.

Independent Research 2026 – Present
Diffusion Language Models

- Working on **diffusion language models (DLMs)**, aiming to improve both their inference speed and generation accuracy, toward practical alternatives to slower autoregressive decoding.
- Investigating decoding and training strategies that reduce the number of denoising steps while preserving output quality, so that generation gets faster without sacrificing accuracy.

Khulna University of Engineering & Technology (KUET)
Undergraduate Thesis Research

- Designed and implemented a **hybrid CNN–Transformer** architecture for 2D polyp image segmentation.
- Combined spatial-context preservation blocks with a Transformer-based decoder and analyzed segmentation performance to understand the contribution of each architectural component.

Selected Projects

Bangla OCR: Progressive Training of a Compact VLM Qwen3-VL · GRPO · vLLM · Jun 2025 – Present

- Built a Bangla document OCR model for handwritten and printed text on a compact vision-language model, using **progressive multi-stage training**: vision-language alignment, supervised OCR, then **reinforcement learning with GRPO** using text-accuracy rewards.
- Built an **automated data engine** that labels unannotated pages by **multi-model consensus voting**, synthesizes camera-captured and degraded pages, and filters noisy labels with **image-to-image self-verification**.
- Served the model on GPUs with **vLLM** for high-throughput batched inference, and added a PyTorch/**ONNX** backend for lighter deployments.

Multimodal RAG Assistant for Scientific Papers Gemini File Search · LangGraph · RAG

- Designed a **multimodal RAG pipeline** over uploaded PDFs that retrieves from text, tables, and figures, and grounds every answer in **cited source passages** to limit hallucination.
- Orchestrated the LLM with a **LangGraph agent** that routes each query to paper summarization, targeted question answering, or citation lookup.
- Built **persistent multi-session chat** and per-user document libraries, for revisiting conversations.

Image-to-Drawing CNC Plotter Arduino · G-code · Stepper motors

- Built a **CNC drawing machine** that takes an image as input and draws the design on paper, using stepper motors for the X and Y axes and a servo to lift and lower the pen.
- Converted input images to **vector toolpaths and G-code**, and programmed the **Arduino** controller to execute them; simulated the circuit in **Proteus** before assembly.

FPGA-Based Arithmetic Logic Unit (ALU) Verilog · Quartus · Altera FPGA

- Designed a multi-bit **ALU** in Verilog supporting arithmetic (add, subtract), logic (AND, OR, XOR, NOT), and shift operations, with **status flags** for carry, zero, and overflow.
- Verified by waveform simulation, synthesized in **Quartus**, and tested on an **Altera FPGA** board.

Technical Skills

Programming & data	Python, MATLAB, Pandas, NumPy
ML & deep learning	PyTorch, Transformers
LLMs & generative AI	Fine-tuning, Unsloth, RAG, LangChain, LangGraph, AI agents
Circuit design	PSpice, LTspice, Proteus, LOGO! Soft Comfort, Altera, Quartus, Arduino IDE
Productivity	Microsoft Word, Excel, PowerPoint, LaTeX

Industrial Visits

- Karnaphuli Hydropower Station
- Cox's Bazar Cable Landing Station
- Radar Station, Cox's Bazar

Honors & Achievements

- **Dean's List Award**, KUET; recognized for a GPA of 3.75 or higher during the 2019–2020 academic session.
- **Technical Education Scholarship**; recognized for ranking among the top 60 students in class.
- **Machine Learning with Python**; completed an intermediate course through the Edge University Activation Program.

Extracurricular Activities

- Owner of **Crochet of Love**, an online business specializing in handmade crafts.
- General member of EEE Makers Hub and KUET Debating Society.
- Presidium member of the RAJUKians of KUET (ROK) Association.
- Organized EEE Day 2023; hosted several university events; performed in university cultural programs.

References

Dr. Kalyan Kumar Halder
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